Python Indexing & Endpoints - Review Guide

MASTER RULE

"Up to, not including"

Key Memory Tricks

First item is [0] not [1]

[start:end] means start to end-1

[-1] is the last item, [-2] is second to last

Last index is always len() - 1

Indexing

Index Numbers

String/List:	P	Y	T	H	O	N
Positive:	0	1	2	3	4	5
Negative:	-6	-5	-4	-3	-2	-1

Getting Single Items

```
word = "PYTHON"

word[0]   # P (first)
word[1]   # Y
word[5]   # N (last)
word[-1]  # N (last)
word[-2]  # O (second to last)
```

Common Error

```
word = "HELLO"  # 5 letters

word[5]   # ERROR! No index 5
word[4]   # O (last = len-1)
word[-1]  # O (easier!)
```

Slicing: Getting Multiple Items

SLICING RULE

[start:end] = start through end-1

word[1:4] Example

```
word = "PYTHON"

 P   Y   T   H   O   N
 0   1   2   3   4   5
     ^--------^
    start    end

word[1:4] = "YTH" (indexes 1, 2, 3 only)
```

Basic Slicing

```
word = "PYTHON"

word[0:3]   # "PYT"
word[1:4]   # "YTH"
word[2:5]   # "THO"
word[0:1]   # "P"
```

Slicing Shortcuts

```
word = "PYTHON"

word[:3]    # "PYT"
word[3:]    # "HON" (3 to end)
word[:]     # "PYTHON"
word[:-1]   # "PYTHO"
word[-3:]   # "HON"
```

Key: No number after colon means "go all the way to the end"
`word[2:]` = index 2 through the end

Negative shortcuts:
`word[:-1]` = everything except last
`word[-3:]` = last 3 items
`word[1:]` = everything except first

range() Function

RANGE RULE

range() works like slicing

Loop from A to B: use `range(A, B+1)`

Code	Numbers You Get	Why
`range(5)`	0, 1, 2, 3, 4	Start at 0 (default), stop before 5
`range(1, 5)`	1, 2, 3, 4	Start at 1, stop before 5
`range(1, 11)`	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	Start at 1, stop before 11 (gives you 1-10!)
`range(0, 10, 2)`	0, 2, 4, 6, 8	Start 0, step by 2, stop before 10

len() Function

LEN() RULE

Last index = len() - 1

len() Example

```
word = "HELLO"

len(word)           # 5
word[0]             # H (first)
word[4]             # O (last)
word[5]             # ERROR!
```

Error: `word[len(word)]` is wrong. Use `word[-1]`

Side-by-Side Comparison

Operation	Start Included?	End Included?
Single index `[i]`	Gets one item	
Slice `[start:end]`	YES	NO
Range `range(start, stop)`	YES	NO

Common Mistakes

1. First item is [0] not [1]

2. [1:4] gives 1,2,3 not 1,2,3,4

3. range(1,10) gives 1-9 not 1-10. Need range(1,11) for 1-10

4. word[len(word)] causes an error. Use word[-1]

Common Patterns

First N items: `my_list[:N]`

Last N items: `my_list[-N:]`

Skip first: `my_list[1:]`

Skip last: `my_list[:-1]`

All indexes: `for i in range(len(my_list)):`

Practice Problems

Problem 1

```
word = "CODING"
# What is word[1:4]?
```

Answer: "ODI" (indexes 1, 2, 3)

Problem 2

```
# Numbers 5 through 10
# What range?
```

Answer: `range(5, 11)`

Problem 3

```
nums = [10, 20, 30, 40, 50]
# What is nums[-2]?
```

Answer: 40

Problem 4

```
text = "PYTHON"
# Last 3 letters?
```

Answer: `text[-3:]` = "HON"

Problem 5

```
word = "HELLO"
# Last valid index?
```

Answer: 4 (len = 5, so 5-1 = 4)

Problem 6

```
word = "PYTHON"
# Skip first and last?
```

Answer: `word[1:-1]` = "YTHO"

Quick Review

THREE THINGS TO REMEMBER

1. First index is 0, last index is len()-1
2. [start:end] and range(start, stop) both exclude the end
3. Negative indexes count from the right: [-1] is last